

QM

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1) The conserved current: Consider a quantum mechanical particle propagating in a given potential and described by the wave function $\Psi(x, t)$. The probability $P(x, t)$ of finding the particle at the position x and the time t is given by

$$P(x, t) = |\Psi(x, t)|^2. \quad (1)$$

Using the one-dimensional Schrödinger equation, show that the probability $P(x, t)$ satisfies the conservation law

$$\frac{\partial P(x, t)}{\partial t} + \frac{\partial j(x, t)}{\partial x} = 0, \quad (2)$$

where the quantity $j(x, t)$ represents the conserved current given by

$$j(x, t) = \frac{\hbar}{2im} \left[\Psi^*(x, t) \left(\frac{\partial \Psi(x, t)}{\partial x} \right) - \Psi(x, t) \left(\frac{\partial \Psi^*(x, t)}{\partial x} \right) \right]. \quad (3)$$

2) Ehrenfest's theorem: Show that

$$\frac{d\langle \hat{p}_x \rangle}{dt} = - \left\langle \frac{\partial V}{\partial x} \right\rangle, \quad (4)$$

a relation that is often referred to as the Ehrenfest's theorem.

3) Conservation of the scalar product: The scalar product between two normalizable wave-functions, say, Ψ_1 and Ψ_2 , which describe a one-dimensional system is defined as

$$\langle \Psi_2 | \Psi_1 \rangle \equiv \int dx \Psi_2^*(x, t) \Psi_1(x, t), \quad (5)$$

where the integral is to be carried out over the domain of interest. Show that

$$\frac{d\langle \Psi_2 | \Psi_1 \rangle}{dt} = 0. \quad (6)$$

4) From the wavefunction to the potential: Consider the one dimensional wave function

$$\psi(x) = A (x/x_0)^n \exp -(x/x_0), \quad (7)$$

where A , n and x_0 are constants. Determine the time-independent potential $V(x)$ and the energy eigen value E for which this wave function is an energy eigen function.

5) Encountering special functions: Solve the Schrödinger equation in a smoothened step that is described by the potential

$$V(x) = \frac{V_0}{2} \left[1 + \tanh \left(\frac{x}{2a} \right) \right], \quad (8)$$

and determine the reflection and the transmission probabilities.

6) The Klein-Gordon equation: Consider the following Klein-Gordon equation governing a wavefunction $\Psi(x, t)$:

$$\frac{1}{c^2} \frac{\partial^2 \Psi(x, t)}{\partial t^2} - \frac{\partial^2 \Psi(x, t)}{\partial x^2} + \left(\frac{\mu c}{\hbar} \right)^2 \Psi(x, t) = 0, \quad (9)$$

where c and μ are constants. Show that there exists a corresponding ‘probability’ conservation law of the form

$$\frac{\partial P(x, t)}{\partial t} + \frac{\partial j(x, t)}{\partial x} = 0, \quad (10)$$

where the quantity $j(x, t)$ represents the conserved current given by

$$j(x, t) = \frac{\hbar}{2i\mu} \left[\Psi^*(x, t) \left(\frac{\partial \Psi(x, t)}{\partial x} \right) - \Psi(x, t) \left(\frac{\partial \Psi^*(x, t)}{\partial x} \right) \right]. \quad (11)$$

1. Express the ‘probability’ $P(x, t)$ in terms of the wavefunction $\Psi(x, t)$.

2. Can you identify any issue with interpreting $P(x, t)$ as the probability?

7) Particle in a box I: Consider a particle of mass m that is confined to a box with its walls at $x = 0$ and $x = a$. It is subject to the additional delta function potential

$$V(x) = V_0 \delta^{(1)}(x - a/2). \quad (12)$$

(a) Write down the energy eigen functions in the domains $0 \leq x \leq a/2$ and $a/2 \leq x \leq a$.

(b) Impose the matching conditions and arrive at the conditions that determine the energy eigen values of the system.

8) The Dirac ‘comb’: Consider a particle propagating in an infinite series of evenly spaced, attractive, Dirac delta function potentials of the following form:

$$V(x) = -a \sum_{n=-\infty}^{\infty} \delta^{(1)}(x - nx_0), \quad (13)$$

where $a > 0$. Such a situation can describe, for instance, the potential encountered by an electron as it traverses along a given direction in a solid. Due to the periodic nature of the potential, one can expect that the energy eigen states satisfy the condition $\psi_E(x + x_0) = e^{iqx_0} \psi_E(x)$ so that $|\psi_E(x + x_0)|^2 = |\psi_E(x)|^2$. Further, to avoid boundary effects, one often imposes the periodic boundary condition

$$\psi(x + Nx_0) = \psi(x), \quad (14)$$

where $N \gg 1$. In such a case, we obtain that

$$\psi(x + Nx_0) = (e^{iqx_0})^N \psi(x) = \psi(x), \quad (15)$$

which leads to

$$(e^{iqx_0})^N = 1, \quad (16)$$

or, equivalently, $q = 2\pi n/(Nx_0)$, where $n = 0, 1, 2, \dots, N$. For large N , q ranges almost continuously between 0 and 2π .

1. By matching the wavefunction and its derivative suitably across one of the delta functions, show that, for bound states wherein $E < 0$, the energy eigen values satisfy the equation

$$\frac{2\kappa x_0 [\cosh(\kappa x_0) - z]}{\sinh(\kappa x_0)} = \frac{2max_0}{\hbar^2}, \quad (17)$$

where $\kappa = \sqrt{-2mE}/\hbar$ and $z = \cos(qx_0)$.

2. Also, show that, for scattering states wherein $E > 0$, the corresponding condition is given by

$$z = \cos(kx_0) - \frac{ma}{\hbar^2 k} \sin(kx_0), \quad (18)$$

where $k = \sqrt{2mE}/\hbar$.

3. Argue that, for $2max_0/\hbar^2 \gg 1$, these conditions lead to a series of energy bands, i.e. an almost continuous range of allowed energy levels, separated by disallowed energy gaps.

9) Quasi-probabilities in phase space: Given a normalized wave function $\Psi(x, t)$, the Wigner function $W(x, p, t)$ is defined as

$$W(x, p, t) = \frac{1}{\pi\hbar} \int_{-\infty}^{\infty} dy \Psi^*(x + y, t) \Psi(x - y, t) e^{2ipy/\hbar}. \quad (19)$$

1. Show that the Wigner function $W(x, p, t)$ can also be expressed in terms of the momentum space wave function $\Phi(p, t)$ as follows:

$$W(x, p, t) = \frac{1}{\pi\hbar} \int_{-\infty}^{\infty} dq \Phi^*(p + q, t) \Phi(p - q, t) e^{-2iqx/\hbar}. \quad (20)$$

Note: Recall that, given the wave function $\Psi(x, t)$, the momentum space wavefunction $\Phi(p, t)$ is described by the integral

$$\Phi(p, t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx \Psi(x, t) e^{-ipx/\hbar}. \quad (21)$$

2. Show that the Wigner function is a real quantity.
3. Show that

$$\int_{-\infty}^{\infty} dp W(x, p, t) = |\Psi(x, t)|^2 \quad \text{and} \quad \int_{-\infty}^{\infty} dx W(x, p, t) = |\Phi(p, t)|^2. \quad (22)$$

10) The energy-time uncertainty principle: Consider a system that is described by the Hamiltonian operator $\hat{H}$.

1. Given an operator, say, $\hat{Q}$, establish the following relation:

$$\frac{d\langle\hat{Q}\rangle}{dt} = \frac{i}{\hbar} \langle[\hat{H}, \hat{Q}]\rangle + \left\langle \frac{\partial\hat{Q}}{\partial t} \right\rangle, \quad (23)$$

where the expectation values are evaluated in a specific state.

2. When $\hat{Q}$ does not explicitly depend on time, using the generalized uncertainty principle, show that

$$\Delta H \Delta Q \geq \frac{\hbar}{2} \left| \frac{d\langle\hat{Q}\rangle}{dt} \right|. \quad (24)$$

3. Defining

$$\Delta t \equiv \frac{\Delta Q}{|d\langle\hat{Q}\rangle/dt|}, \quad (25)$$

establish that

$$\Delta E \Delta t \geq \frac{\hbar}{2}, \quad (26)$$

and interpret this result.

11) A three-dimensional vector space: Consider a three-dimensional vector space spanned by the orthonormal basis $|1\rangle$, $|2\rangle$ and $|3\rangle$. Let two kets, say, $|\alpha\rangle$ and $|\beta\rangle$ be given by

$$|\alpha\rangle = i|1\rangle - 2|2\rangle - i|3\rangle \quad \text{and} \quad |\beta\rangle = i|1\rangle + 2|3\rangle. \quad (27)$$

1. Construct $\langle\alpha|$ and $\langle\beta|$ in terms of the dual basis, i.e. $\langle 1|$, $\langle 2|$ and $\langle 3|$.
2. Find $\langle\alpha|\beta\rangle$ and $\langle\beta|\alpha\rangle$ and show that $\langle\beta|\alpha\rangle = \langle\alpha|\beta\rangle^*$.
3. Determine all the matrix elements of the operator $\hat{A} = |\alpha\rangle\langle\beta|$ in this basis and construct the corresponding matrix. Is the matrix hermitian?

12) A two level system: The Hamiltonian operator of a certain two level system is given by

$$\hat{H} = E(|1\rangle\langle 1| - |2\rangle\langle 2| + |1\rangle\langle 2| + |2\rangle\langle 1|), \quad (28)$$

where $|1\rangle$ and $|2\rangle$ form an orthonormal basis, while E is a number with the dimensions of energy.

1. Find the eigen values and the normalized eigen vectors, i.e. as a linear combination of the basis vectors $|1\rangle$ and $|2\rangle$, of the above Hamiltonian operator.
2. What is the matrix that represents the operator $\hat{H}$ in this basis?

13) Matrix elements for the harmonic oscillator: Let $|n\rangle$ denote the orthonormal basis of energy eigen states of the harmonic oscillator. Determine the matrix elements $\langle n|\hat{x}|m\rangle$ and $\langle n|\hat{p}_x|m\rangle$ in this basis.

14) Coherent states of the harmonic oscillator: Consider states, say, $|\alpha\rangle$, which are eigen states of the annihilation (or, more precisely, the lowering) operator, i.e.

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle, \quad (29)$$

where α is a complex number.

Note: The state $|\alpha\rangle$ is called the coherent state.

1. Calculate the quantities $\langle\hat{x}\rangle$, $\langle\hat{x}^2\rangle$, $\langle\hat{p}_x\rangle$ and $\langle\hat{p}_x^2\rangle$ in the coherent state.
2. Also, evaluate the quantities Δx and Δp_x in the state, and show that $\Delta x \Delta p_x = \hbar/2$.
3. Like any other general state, the coherent state can be expanded in terms of the energy eigen states $|n\rangle$ of the harmonic oscillator as follows:

$$|\alpha\rangle = \sum_{n=0}^{\infty} c_n |n\rangle. \quad (30)$$

Show that the quantities c_n are given by

$$c_n = \frac{\alpha^n}{\sqrt{n!}} c_0. \quad (31)$$

4. Determine c_0 by normalizing $|\alpha\rangle$.
5. Upon including the time dependence, show that the coherent state continues to be an eigen state of the lowering operator $\hat{a}$ with the eigen value evolving in time as

$$\alpha(t) = \alpha e^{-i\omega t}. \quad (32)$$

Note: Therefore, a coherent state *remains* coherent, and continues to minimize the uncertainty.

6. Is the ground state $|0\rangle$ itself a coherent state? If so, what is the eigen value α ?
- 15) Feynman-Kac formula: Given that a quantum mechanical particle is at the location x at time t , show that the probability amplitude for finding the particle at the location x' at a later time t' can be expressed as

$$K(x', t'|x, t) = \langle x', t'|x, t \rangle = \sum_{\text{all } n} \psi_n(x') \psi_n^*(x) e^{-iE_n(t'-t)/\hbar}, \quad (33)$$

where $\psi_n(x)$ denote the energy eigen states corresponding to the energy eigen values E_n and the sum over n represents either the sum or a suitable integral in the case of a continuum of states over all the energy eigen values. Evaluate the quantity $\langle x', t'|x, t \rangle$ for the case of a free particle.

Note: The above expression for the probability amplitude is known as the Feynman-Kac formula. The probability amplitude $\langle x', t'|x, t \rangle$ can be given a so-called path integral interpretation, i.e. it can be expressed as a sum over all paths with the probability amplitude for each path given the same weightage, being proportional to $\exp\{iS[x(t)]/\hbar\}$, with $S[x(t)]$ denoting the corresponding classical action.

- 16) The raising and lowering angular momentum operators: As we have discussed, the raising and lowering angular momentum operators L_+ and L_- change the value of the z -component of angular momentum, viz. the eigen value m (corresponding to the operator L_z) by one unit, i.e.

$$L_{\pm} f_l^m = A_l^m f_l^{m\pm 1}, \quad (34)$$

where A_l^m are constants, while f_l^m are simultaneous eigen functions of the operators L^2 and L_z . What are A_l^m , if f_l^m are normalized eigen functions?

- 17) Larmor precession: Consider a charged, spin 1/2 particle that is at rest in an external and uniform magnetic field, say, $\mathbf{B}$, that is oriented along the z -direction, i.e. $\mathbf{B} = B \hat{k}$, where B is a constant. The Hamiltonian of the particle is then given by

$$\hat{H} = -\gamma B \hat{S}_z, \quad (35)$$

where γ is known as the gyromagnetic ratio of the particle.

1. Determine the most general, time dependent, solution that describes the state of the particle.
2. Evaluate the expectation values of the operators $\hat{S}_x$, $\hat{S}_y$ and $\hat{S}_z$ in the state.
3. Show that the expectation value of the operator $\hat{\mathbf{S}} = \hat{S}_x \hat{i} + \hat{S}_y \hat{j} + \hat{S}_z \hat{k}$ is tilted at a constant angle with respect to the direction of the magnetic field and precesses about the field at the so-called Larmor frequency $\omega = \gamma B$.
- 18) Spin- $\frac{3}{2}$ particle: Consider a particle with spin $\frac{3}{2}$.

1. What are the eigen values and eigen states of the $\hat{S}_z$ operator for the system?

2. Determine the effects of the operators $\hat{S}_+$ and $\hat{S}_-$ on these eigen states.
3. Construct the matrices describing the operators $\hat{S}_+$ and $\hat{S}_-$.
4. Obtain the matrix describing the operator $\hat{S}_x$.